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FIN 43900 — AI Finance Applications · Fall 2026 Syllabus

Generated copy for offline reading. The course syllabus of record is on Brightspace (Syllabus tab). If this PDF and Brightspace disagree, Brightspace wins.

Course Information

(Meeting times/rooms auto-populate for the CRN being edited. Because the two sections are taught as one course, add the block below so students in either section see both.)

FIN 43900 — AI Finance Applications is taught as one combined course across two sections. Both sections follow the same syllabus, schedule, assignments, rubrics, and deadlines.

Section	CRN	Meets	Room
001	12506	Tuesday & Thursday, 1:30–2:45 pm	Lilly Hall G-420
002	12507	Tuesday & Thursday, 4:30–5:45 pm	Rawls Hall 2070

Term: August 25 – December 10, 2026. Credits: 3.

Instructor(s) Contact Information

(Instructor name and email auto-populate and are editable. Add the rest.)

Dr. Xinde “Cinder” Zhang — Instructor Office: Krannert Building, Room 432 Email: zhan6028@purdue.edu Office hours: By appointment

Beckett Thomas — Teaching Assistant Email: thom1214@purdue.edu Office hours: By appointment

Edward Ter-Hovhannisyanyan — Teaching Assistant Email: eterhovh@purdue.edu Office hours: By appointment

How to reach us. Include “FIN 43900” in the subject line and copy both teaching assistants on every email — messages sent without them may not receive a timely reply. Expect a response within 48 hours, Monday through Friday. Questions about course content belong in the Brightspace Q&A discussion forum, where the whole class benefits from the answer; email is for personal matters. To meet, request an appointment by email.

Course Description

Auto-populated from Purdue’s systems and NOT editable by the instructor (verified in the live tool 2026-08-12). Recorded here so the official text is known; do not attempt to change it.

Credit Hours: 3.00. Applied problem-solving course developing AI-augmented finance competency. Students tackle real corporate finance and investment challenges using current market data and AI tools. The DRIVER framework provides systematic methodology for AI collaboration while maintaining financial judgment. Three major projects build portfolio-ready applications across finance domains.

Prerequisites: Undergraduate level MGMT 31000, minimum grade C–; and undergraduate level MGMT 41100, minimum grade D–.

Course Learning Outcomes

Auto-populated and NOT editable. The official catalog list has SIX outcomes (verified in the live tool 2026-08-12), not the eight in the Spring 2026 syllabus:

By the end of the course, you will be able to:

1. Apply the DRIVER framework systematically to solve complex finance problems with AI collaboration.
2. Orchestrate AI tools effectively for financial data analysis, modeling, forecasting, and decision support.
3. Validate AI-generated outputs using financial domain knowledge, business judgment, and sanity testing.
4. Build practical financial applications that solve classes of problems using current market data.
5. Integrate corporate finance and investment concepts in comprehensive analytical solutions.
6. Communicate solutions professionally through structured video presentations and documentation.

 **Spring 2026 outcomes 7 and 8 are NOT in the catalog** — “Exercise professional judgment...” and “Analyze real-time financial data...”. They exist only in the instructor’s own syllabus. They are not lost: the tool’s help text states that further explanation of these outcomes may be added to the optional block below, which is exactly where they belong.

Course Learning Outcomes (continued)

In addition to the University outcomes above, this course develops two further capabilities:

7. Exercise professional judgment in determining appropriate AI use, maintaining human centrality in financial decisions.
 8. Analyze real-time financial data to make evidence-based recommendations on current business situations.
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Teaching Philosophy

AI makes answers cheap; it does not make understanding, judgment, or growth automatic. The scarce work shifts to choosing questions worth answering, deciding what to trust and do, and developing the capacity to own the consequences. My role is not to hand you finished answers or complete code. I design problems, tools, and feedback loops that let you build, test, fail, explain, and improve. That is why this course emphasizes the D stage of DRIVER: discover and define the right problem before AI accelerates the wrong one. We build to learn—not learn merely to build. The goal is not to outperform AI. It is to use AI as a studio for developing your own understanding, judgment, and capacity to act.

Growth and wisdom matter deeply, but I do not grade them as personality traits or ask you to perform transformation. I grade visible acts of judgment: asking better questions, surfacing assumptions, calibrating confidence to evidence, testing independently, revising after failure, naming limitations, and transferring what you learned to a new problem. You are not penalized for where you begin or rewarded for merely claiming growth.

How to Succeed in this Course

Start with D: discover and define. Before you build, write down the problem, who it is for, what you know, what you need to learn, the real constraints, and what success looks like. A polished solution to the wrong problem is still failure.

Use the best available AI models and harnesses—but do not confuse capability with reliability. Working at the frontier reveals both AI's capacity and its shortcomings. Treat every output as an experiment: prompt, run, inspect, challenge, revise, and document. Test it against known answers, reasonable ranges, edge cases, and a second independent model. Practice through many small cycles; failure is data. Never submit work you cannot explain. Success is turning an AI-produced result into your own defensible financial judgment.

Build locally from the beginning. Use AI to help you plan, write, debug, and test code, but you must be able to run that code on your own computer, inspect errors, install dependencies, manage a multi-file project, and turn analysis into a working web

application. Use GitHub as the durable home for your code and development history, not as a last-minute upload site. Google Colab is an alternative for notebook-based work; it is not a substitute for the required local environment or for testing a web application locally.

Learning Resources, Technology & Texts

REQUIRED: OpenAI ChatGPT Plus subscription — must include Codex access

REQUIRED: Google AI Pro subscription — must include Gemini Pro models

No textbook purchase required — all readings and materials provided on Brightspace

Both AI platforms are required for the full semester. This course is built around working across two independent AI systems: you will use each to cross-examine the other, which is central to the validation skill this course teaches and to how your projects are assessed.

Subscribe to the correct tier. Free accounts on either platform will not carry you through this course. Before the first lab, sign in to each platform and confirm you can reach Codex on OpenAI and Gemini Pro models on Google. If you cannot, you are on the wrong tier; fix it before the first assignment is due, not after. Subscription tier names change periodically; what matters is access to the two capabilities named above. You may use additional AI tools at your own discretion — only these two are required, and all AI use must be disclosed in your submissions.

Reference materials, DRIVER framework guides, templates, problem statements, and data sources are provided through Brightspace. Optional reference texts are available in the DRIVER series at cinderzhang.github.io.

Computing and coding environment: Bring a personal laptop and charger to the first class; create or recover the accounts named in Start Here, but do not guess at package versions or buy an unlisted tool. Week 1 supplies the tested Python, package-manager, editor/harness, Git, browser, and GitHub setup path and records an actual readiness result before later project work depends on it. Students with a blocker preserve the exact symptom and use the published Colab/offline route while the instructional team assigns an owner and next action; a setup failure is not represented as success. The

eventual project environment must run Python scripts, isolated packages, multi-file work, Git/GitHub, and a local web application. Google Colab is an acceptable notebook/troubleshooting route but does not by itself verify that later local-web-app requirement. The required-technology access/equity route remains a release gate and must be published before students are held responsible for equipment or connectivity beyond the announced baseline. It must close before Start Here becomes learner-visible and no later than **Thursday, August 20 at 11:59 p.m. ET**. If it does not, a non-possession posture activates automatically on August 21: dual-platform work uses accessible full-chain output packets and at least one student-authored follow-up probe executed through a free or course-run session, while local work uses the loaner/Colab/offline route. One grading standard applies; no criterion may require evidence available only from an individually purchased tier, and no student is graded on possession of a paid tier or capable personal device. The Start Here object **AI Access — No Purchase Required and No-Cost Comparison Route** supplies the executable route and Week 1 dual-system packet.

Video recording: Screen recording with webcam and clear audio, at 720p or better. Kaltura Capture is available at no cost through Purdue; Zoom recording or other tools are acceptable.

Data sources — all free: SEC EDGAR company filings and API, market data APIs, and economic indicators from FRED, the Bureau of Labor Statistics, and the Census Bureau. Course-specific datasets are provided when needed.

Assignments

Total: 1,400 points. Both sections follow identical assignments and deadlines.

Component	Points	Due
In-class labs (best 8 Tuesday + best 8 Thursday, at 25 points each)	400	In class, Aug 25–Nov 17, excluding studio weeks
Project 1 — Corporate Finance Application	300	Monday, October 19
Project 2 — Investment Management Application	300	Monday, November 23
Capstone — Integrated Financial Analysis System	400	Thursday, December 10

All project deadlines are 11:59 PM Eastern. This syllabus governs course points, grading, and policies. Brightspace assignment pages govern detailed specifications, required files, and submission procedures; announced changes will follow the syllabus-change policy.

In-class labs. Twenty live labs are offered at 25 points each: 10 Tuesday completion labs and 10 Thursday merit-based labs. Your best eight Tuesday scores and best eight Thursday scores count, for 400 points total. This split ensures that neither attendance alone nor merit alone can replace the other. Project studio weeks (7 and 12) are open Q&A and consultation sessions and have no lab checkout.

The two drops in each category are an absence and recovery buffer, not replacement learning. Later sessions deliberately re-probe earlier capabilities on changed inputs; skipping one may remove that practice opportunity even when the score is ultimately dropped.

Each learner submits only for themselves through the dated individual Brightspace lab checkout quiz before leaving class. The checkout records the Lab Date (), teammates/learning partners, financial problem or decision, personal contribution, changed-input test, data and evidence, validation, and—only when the session produces a durable artifact—an optional repository or output link. Teammate names help reconstruct collaboration but never grant or deny attendance credit for another student. Tuesday labs earn 25 points when the authenticated Brightspace attempt is on time, contains the required work record, and includes the in-person attendance declaration and truth attestation; otherwise they earn 0. Thursday labs earn 0–25 points across problem and decision definition, data and evidence integrity, validation and

robustness, financial judgment, and explanation and transfer. Reflection, confidence, effort, personality, and claimed growth or wisdom are not graded. The submitted Brightspace attempt and receipt are the authoritative checkout record. Lab checkouts are never accepted late, and no make-up lab or substitute assignment will be offered except when an equal-credit alternative is required by University policy or an approved accommodation.

Dropping a cold-run or readiness checkout from the counted lab grade does not waive the same evidence in a major project. An ungraded readiness-feedback path remains available before each project deadline; it earns no lab points and does not recreate or replace a missed checkout.

Projects. Each project delivers a locally runnable financial web application or application component plus a professional evidence package. The same submission architecture applies to every student on Project 1, Project 2, and the Capstone: a timestamped Edition A baseline checkpoint declared before permitted project-specific AI work; an after-AI Edition B research-evolution brief; an accessible GitHub repository; a validation and AI-use record; an investment-committee memo or decision deck; and three short videos covering research evolution, technical ownership, and investment judgment. Detailed specifications, filenames, video limits, and submission procedures are published on the Brightspace assignment page.

The submitted files and setup instructions must allow the work to be run and tested on a local computer. The repository contains the runnable source, dependency file, README setup instructions, meaningful development history, and visible executed output. Never commit passwords, API keys, tokens, credentials, or private data; use environment variables and an appropriate `.gitignore`. DRIVER is taught as a problem-solving scaffold, but students are not graded on adherence to the framework itself. Project 1 is scaffolded; Project 2 expects greater independence; the Capstone expects professional-level integration across corporate finance and investment management.

Every project is assessed on the same six criteria:

Criterion	Weight
Problem definition and decision usefulness	15%
Data integrity and traceability	15%
Financial soundness and judgment	20%
Technical execution and usability	15%
Validation, sensitivity, and reproducibility	25%
Professional communication	10%

The published major-project analytic rubric supplies the authoritative performance levels, descriptors, 300/400-point mappings, calibration procedure, no-double-deduction examples, and visible-output cap examples. Assignment pages use copies of that one source.

What the criteria mean. Data integrity includes the correct company or security, reporting period and as-of date, source, units, currency, definitions, transformations, missing-data treatment, and reconciliation to authoritative data. Financial soundness includes correct finance principles, internally consistent assumptions, accounting and cash-flow treatment, time value of money, defensible discount rates, and economically plausible constraints. For example, a perpetual growth rate must not exceed a defensible long-run nominal growth rate for the relevant economy; no company can outgrow the economy forever. Validation includes independent checks, known-answer comparisons, sensitivity and stress testing, edge cases, and reproducibility. Technical execution covers whether the application runs reliably and is usable. A polished application cannot compensate for materially inaccurate data or financially impossible assumptions. These criteria grade evidence in the work, not personality, confidence, or claimed transformation.

Every project submission requires: the unchanged timestamped Edition A baseline; the Edition B research-evolution brief ending with what changed and why; a reproducible GitHub repository and visible executed output; a validation and named AI-use record; a concise investment-committee memo or decision deck; and three separately titled videos with transcripts: (1) research before and after project-specific AI, with course/own/peer/AI change attribution, (2) a detailed codebase and validation

walkthrough, and (3) a formal buy-side investment-committee presentation. The Brightspace assignment page specifies the duration cap and exact manifest. These artifacts provide evidence for the rubric criteria; their conformity to DRIVER is not graded separately. Work submitted without visible executed output receives at most 50% credit — the result is the deliverable, not the code alone.

Late project submissions. The first late project submission in the course may be accepted for up to four calendar days, with 20% of the earned score deducted for each day or fraction of a day late. At five days late, it earns zero. Any second late project submission automatically earns zero. Technical problems do not stop the late clock; submit early and verify the Brightspace receipt. University-approved absences and DRC accommodations are handled as required by University policy.

Grading Scale

Grade	Approximate reference range	Grade	Approximate reference range
A	93–100%	C	73–76%
A–	90–92%	C–	70–72%
B+	87–89%	D+	67–69%
B	83–86%	D	63–66%
B–	80–82%	D–	60–62%
C+	77–79%	F	Below 60%

The percentage ranges above are planning estimates, not guaranteed or fixed cutoffs. Under Daniels School of Business undergraduate grading guidance, this course targets a mean GPA of approximately 3.0, with a target distribution of 20–25% A/A–, 60–70% B+/B/B–, and 10–20% C+ and below. Final letter grades depend on relative performance and may be adjusted upward or downward based on class performance and the DSB grading guide. A numerical total or rounding convention alone does not guarantee a particular letter grade.

Regrade requests must be submitted in writing within seven days of a grade being released and must identify the specific rubric criterion in dispute.

Attendance Policy

Attendance is not graded separately. Twenty class meetings between August 25 and November 17 include an individual lab checkout: 10 Tuesday completion labs and 10 Thursday merit-based labs. Your best eight Tuesday scores and best eight Thursday scores count. The other ten meetings — the Project 1 and Project 2 studio weeks (Oct 6, Oct 8, Nov 10, Nov 12) and the six meetings after November 17 — are Q&A, studio, and consultation sessions with no lab checkout and no lab points.

Each learner must submit only for themselves before leaving the session through the dated Brightspace checkout, record the **Lab Date** (mm/dd/yyyy), declare their in-person attendance, and truthfully record their own participation, teammates/learning partners, contribution, tests, and evidence. A repository/output link is optional and appears only when the session produces a durable artifact. Being named on another student's checkout has no effect on your grade. A mismatch between records is a review flag, not an automatic attendance decision or misconduct finding. Because each lab depends on live work with classmates, there are no late lab submissions or ordinary make-up labs. The two-drop buffers in each category provide general flexibility for missed or weaker lab sessions; they are not represented as a substitute for an approved accommodation.

University-approved absences and Disability Resource Center accommodations will be handled as required by University policy, but not by recreating a past interactive lab. If an approved accommodation or official absence requires a different response, contact me promptly so I can coordinate the documented course-level response with the Office of the Dean of Students or the Disability Resource Center. That plan may change how the affected checkout is counted or provide an approved equal-credit alternative. Do not send medical documentation to the instructional team.

Be realistic about fit. This course is built on live, in-class work, and beyond the built-in buffers there are no make-ups; sustained absence is very difficult to recover from. If your semester schedule will not allow regular Tuesday/Thursday attendance, take that seriously while the registrar's drop/add period is open — a course whose grade lives in the classroom may not be the right choice this term, and deciding early protects your

GPA and your tuition. If your absences stem from circumstances the University recognizes — illness, an approved accommodation, an official absence — use the routes above with the Office of the Dean of Students or the Disability Resource Center early, and they will be handled as University policy requires.

Course Schedule

(Generated from MASTER-SCHEDULE.md. All sessions meet Tuesday and Thursday. No class Tuesday, October 13 (Fall Break) or Thursday, November 26 (Thanksgiving Break). Every class meeting is an interactive working session; this schedule lists topics and major deadlines only.)

Weeks	Dates	Topic	Major milestone
1	Aug 25, Aug 27	Technology readiness, DRIVER, and company screening	—
2–4	Sep 1, Sep 3, Sep 8, Sep 10, Sep 15, Sep 17	Valuation and deal analysis	—
5–6	Sep 22, Sep 24, Sep 29, Oct 1	Credit risk assessment and transaction screening	—
7	Oct 6, Oct 8	Project 1 studio — open Q&A/consultation (no lab checkout)	—
8	Oct 15 (<i>Fall Break Oct 12–13</i>)	Portfolio construction opens	Project 1 due Mon Oct 19
9	Oct 20, Oct 22	Portfolio construction and optimization	—
10–11	Oct 27, Oct 29, Nov 3, Nov 5	Alternative data and market analysis	—
12	Nov 10, Nov 12	Project 2 studio — open Q&A/consultation (no lab checkout)	—
13	Nov 17, Nov 19	Applied decision-system studio — student-selected track (<i>final lab checkout Tue Nov 17; no checkout Nov 19</i>)	Project 2 due Mon Nov 23
14	Nov 24	Applied decision-system studio continues (no lab checkout)	No class Thu Nov 26
15	Dec 1, Dec 3	Integrated capstone development (<i>no lab checkouts</i>)	—
16	Dec 8, Dec 10	Capstone consultation studios (<i>no lab checkouts</i>)	Capstone due Thu Dec 10

Twenty meetings through November 17 include a graded lab checkout. Ten meetings run none: the project studio weeks (Oct 6, Oct 8, Nov 10, Nov 12) and the six meetings after November 17 (Nov 19, Nov 24, Dec 1, Dec 3, Dec 8, Dec 10). Tuesday labs are primarily completion based; Thursday labs are primarily merit based. Tuesday points measure the authenticated required work record, not physical presence by itself. The final six class meetings are project and capstone sessions. Deadlines are published here and in the Brightspace calendar and will not move earlier.

Academic Integrity

Purdue prohibits dishonesty in connection with any University activity. Cheating, plagiarism, and knowingly furnishing false information to the University are examples of dishonesty (Part 5, Section III-B-2-a, University Regulations).

Specific to this course:

You are responsible for every submitted claim, assumption, output, and material code path, regardless of whether AI produced it. Instructors may request a live demonstration or oral explanation to assess authorship and understanding. Inability to explain the work affects the relevant grading criteria and may prompt an academic-integrity review, but it is not by itself an automatic misconduct finding; any misconduct determination follows Purdue procedures.

Undisclosed AI use is an academic integrity violation. Disclosed AI use that contains unedited AI text or factual errors is a quality problem and is graded down, not referred — the difference is disclosure.

Collaboration is encouraged during in-class labs. Individual project submissions must be your own work; sharing project deliverables between students is not permitted.

Knowingly submitting a lab checkout for a session you did not attend, submitting on another student's behalf, or falsifying teammates/learning partners, contribution, the in-person attendance declaration, or evidence is also prohibited. A discrepancy is reviewed rather than treated as automatic proof of misconduct.

AI Policy

AI use is required in this course, not merely permitted. Learning to direct AI toward correct, verifiable financial analysis is the central skill being taught and assessed.

The same requirements apply to every student on every assignment:

Disclose. Every submission states which AI access path you used and for what. Record only model identifiers the interface actually exposed. If you used a published packet or a course-run response, name that path and do not claim its collection tools as your own use. A vague or unnamed access path does not satisfy the disclosure element.

Document. Every project includes an AI-use and validation record — key prompts, what the AI produced, and how you verified it — plus a reflection on what AI did well, where it struggled, and what you checked independently.

Validate. Every AI-produced result is a hypothesis until you test it. Check it against a known answer, check it for reasonableness, check the edge cases, and check for the failure modes specific to AI, such as fabricated sources and confidently wrong figures. This course requires comparison of two independent system outputs—through packets, available free interfaces, a funded Purdue route if announced, or course-run responses—partly so you can cross-examine one with the other.

Own it. You are responsible for everything you submit. AI is a collaborator, not an author, and human judgment stays central to every financial decision you present.

Nondiscrimination Statement

(Use Purdue's standard statement as provided in the template. Do not paraphrase.)

Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. Purdue University prohibits discrimination against any member of the University community on the basis of race,

religion, color, sex, age, national origin or ancestry, genetic information, marital status, parental status, sexual orientation, gender identity and expression, disability, or status as a veteran.

Notes for entry

Word, which is a common source of accessibility-checker failures.

- Keep tables simple with a header row. Do not nest tables.
- Make link text descriptive — never “click here.”
- Watch the live accessibility checker as you go and clear every flag before submitting.
- Two items still marked TBD: office hours, and teaching assistants. Both must be filled before submission.